THE QUASI-COHERENT PRESHEAF ASSOCIATED TO A MODULE SATISFIES THE SHEAF CONDITION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The spanned statement cites the chapter's own equation labelling the functor in question and the definition of a tau-covering, both delivered as navigation stubs.

Editorial note. No bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Every Zariski, etale, smooth, syntomic or fppf covering is an fpqc covering. Topologies, Lemmas topologies-lemma-zariski-etale, topologies-lemma-zariski-etale-smooth, topologies-lemma-zariski-etale-smooth-syntomic, topologies-lemma-zariski-etale-smooth-syntomic-fppf, topologies-lemma-zariski-etale-smooth-syntomic-fppf-fpqc, invoked at proof lines 2386-2395. Reduces the sheaf condition for all five topologies to the fpqc case.
- Effectivity of fpqc descent for quasi-coherent modules. Same chapter, Proposition proposition-fpqc-descent-quasi-coherent, invoked at proof lines 2405-2407. Descends the compatible family of maps to a unique map O_T -> f^* F.

Source reference. Original equation 03DS: sources/stacks-descent/pinned-original-source.tex, source lines 2368–2368. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original proposition 023T: sources/stacks-descent/pinned-original-source.tex, source lines 1801–1801. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0216: sources/stacks-chow/tags.txt, source lines 2659–2659. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0220: sources/stacks-chow/tags.txt, source lines 2689–2689. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0226: sources/stacks-chow/tags.txt, source lines 2695–2695. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 021N: sources/stacks-chow/tags.txt, source lines 2676–2676. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 022C: sources/stacks-chow/tags.txt, source lines 2701–2701. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_S$-module. Let $\tau \in \{Zariski, étale, smooth, syntomic, fppf, fpqc\}$. The functor defined in (03DS) satisfies the sheaf condition with respect to any τ-covering $\{T_i \to T\}_{i\in I}$ of any scheme T over S.*

Proof. For $\tau \in \{Zariski, étale, smooth, syntomic, fppf\}$ a τ-covering is also a fpqc-covering, see the results in Topologies, Lemmas 0216, 0220, 0226, 021N, and 022C. Hence it suffices to prove the theorem for a fpqc covering. Assume that $\{f_i : T_i \to T\}_{i\in I}$ is an fpqc covering where $f : T \to S$ is given. Suppose that we have a family of sections $s_i \in \Gamma(T_i, f_i^* f^*\mathcal{F})$ such that $s_i|_{T_i\times_T T_j} = s_j|_{T_i\times_T T_j}$. We have to find the correspond section $s \in \Gamma(T, f^*\mathcal{F})$. We can reinterpret the s_i as a family of maps $\varphi_i : f_i^*\mathcal{O}_T = \mathcal{O}_{T_i} \to f_i^* f^*\mathcal{F}$ compatible with the canonical descent data associated to the quasi-coherent sheaves $\mathcal{O}_T$ and $f^*\mathcal{F}$ on T. Hence by Proposition 023T we see that we may (uniquely) descend these to a map $\mathcal{O}_T \to f^*\mathcal{F}$ which gives us our section s. □